mNGS

Lab setup and sample preparation

Key points to learn in this session

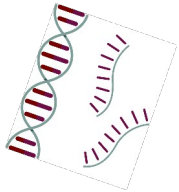
- Sequencing technologies
- Lab setup & sample collection
- Nucleic acid extraction
- Extraction QC

mNGS pipeline

Study design



Nucleic acid extraction



Library prep



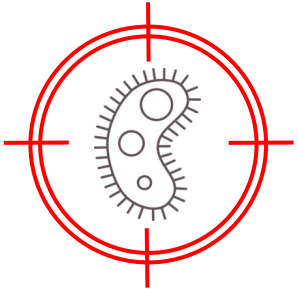
Sequencing



Data analysis



Pathogen ID

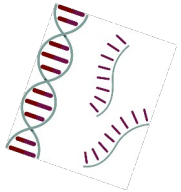


mNGS pipeline

Study design



Nucleic acid extraction



Library prep



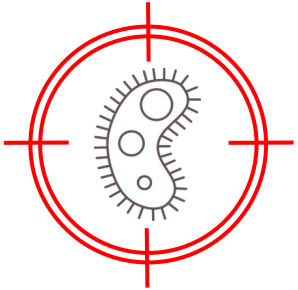
Sequencing



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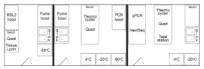


Depends on sequencing platform
Happens in lab space

mNGS pipeline



Let’s talk about sequencing platforms and lab spaces!



Lab setup

Important considerations for setting up a lab & collecting samples for mNGS

Concern: Working with infectious agents

Solution: inactivate with DNA/RNA shield



Concern: Preserving sample/nucleic acid integrity

Solution: Maintain cold chain or use DNA/RNA shield, minimize freeze-thaws, prevent RNAse contamination

Concern: Cross contamination - **avoid PCR amplicon contamination at all cost**

Solution: Physical separation of rooms (Pre-PCR & Post-PCR)

Concern: Environmental conditions (humidity, temperature, power, internet)

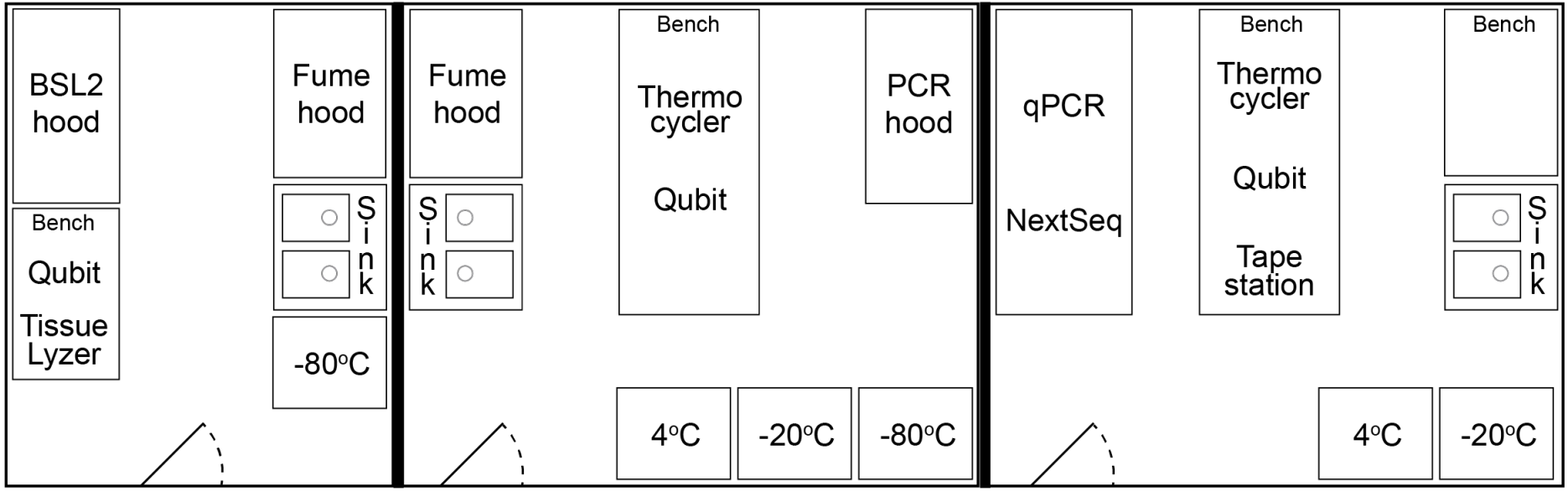
Solution: Air conditioning & air filter units in the lab, UPS, running overnight

Lab setup



*It is **critical** to have separate pre- and post-PCR lab spaces!*

Lab setup



BSL-2 Pre-PCR

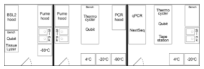
- **No PCR allowed**
- xNA extractions of infectious samples

BSL-1 Pre-PCR

- **No PCR allowed**
- xNA extractions
- Lib prep up to setting up the USER/Q5 mastermix (no PCR)

BSL-1 Post-PCR

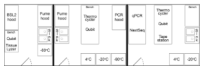
- **Lib prep from barcoding PCR**
- Lib QC
- Lib amplification if needed
- Sequencing



Lab setup

Extra lab precautions:

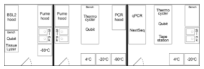
- Sticky floor pads at the entrance of Pre-PCR rooms, changed weekly
- Ice buckets color coded to the lab spaces - either Pre-PCR or Post-PCR only - no mixing
- Rotation of index primers so the same indices are never used in the same month
- Reagents like H_2O, SPRI beads, DNAse, Proteinase K, etc are aliquoted into single-use tubes
- Pipets inside the PCR hoods are never taken out. All hoods are sprayed down before and after use with 70% ethanol and RNAseZap
- Reagents are not shared between the different rooms
- PCR products never enter the Pre-PCR rooms nor samples with high nucleic acid quantities (i.e.: stool, tissues, bacterial cultures, bacterial colonies)



Lab setup

Other considerations:

- Power: Uninterruptible power supply (UPS) or surge protector
- Internet connection
- Environmental conditions, e.g. for iSeq:
  - Dust issues: Pollution degree rating of II (non-conductive pollution only, rare conductive pollution)
  - Ventilation: Up to 2048 BTU/hr @ 600W. Buy extra air filters for the iSeq.
  - Humidity: Non-condensing 20-80% Relative Humidity
  - Temperature: 15°C to 30°C (22.5°C ± 7.5°C)



Lab setup

Lab procedures by location

BSL2 Pre-PCR Biosafety Cabinet

- Sample inactivation in DNA/RNA Shield
- Nucleic acid extraction (if sample is not deactivated)

BLS2-Pre-PCR Room

- Storage of original samples at -80C

BSL1 Pre-PCR PCR hood

- Nucleic acid extraction (if sample is deactivated)
- Nucleic acid quantification/quality control:
  - Aliquots can be made in Pre-PCR and run in Post-PCR
- DNA/RNA library prep: Fragmentation to setting up the USER/Q5 master mix

BSL1 Post-PCR PCR hood

- Post-PCR SPRI clean-up

BSL1 Post-PCR Bench

- PCR reaction using USER/Q5 master mix
- Library quantification & quality control
- Library pooling
- qPCR Machine for additional library amplification
- Illumina sequencing
- Storage of NGS libraries -20C

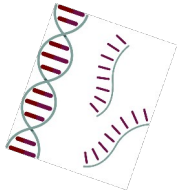
Questions?

mNGS pipeline

Study design



Nucleic acid extraction



Library prep



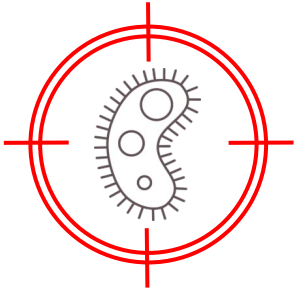
Sequencing



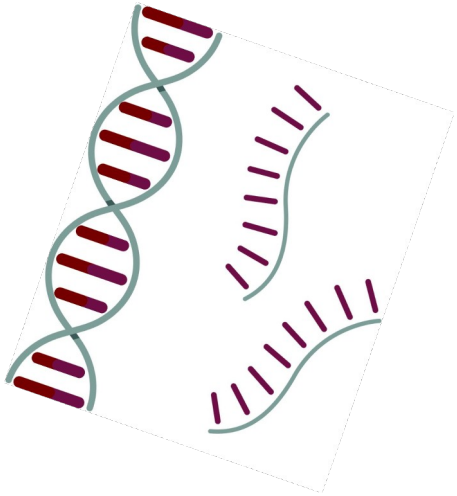
Data analysis

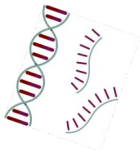


Pathogen ID



Nucleic acid extraction





xNA extraction

Key points:

✓ Minimize cross-contamination

✓ Preserve sample quality

- Before extraction:
  - Spin down to remove host cells (extract supernatant)
- Choose type:
  - DNA:
    - Detect bacteria, parasites, fungi, DNA viruses
  - RNA:
    - All the above + RNA viruses (!)
    - Requires optimal sample quality, degrades very fast
    - Keep cold at all times!
  - TNA (total nucleic acid):
    - DNA + RNA
    - Requires DNAse I step for RNA-mNGS
- Column vs bead-based extraction

xNA extraction

Zymo DNA+RNA extraction kits



Quick-DNA/RNA Microprep Plus

50 preps for $399

Column-based



Quick-DNA/RNA Miniprep Plus

50 preps for $399

Column-based



Quick-DNA/RNA Pathogen Magbead

96 preps for $293

Bead-based

Requires extra:

- ✔ Magnet stand (CZB Engineers)
- ✔ β-mercaptoethanol

Extra reagents to buy:

- ✔ Molecular grade ethanol (200 proof) for washes
- ✔ Isopropanol to add to concentrated wash buffers
- ✔ DNase I included in regular quick-DNA/RNA kits, but not in Pathogen kits
  - Available separately from Qiagen: RNase-Free DNase Set, $512 for 250 samples

xNA extraction

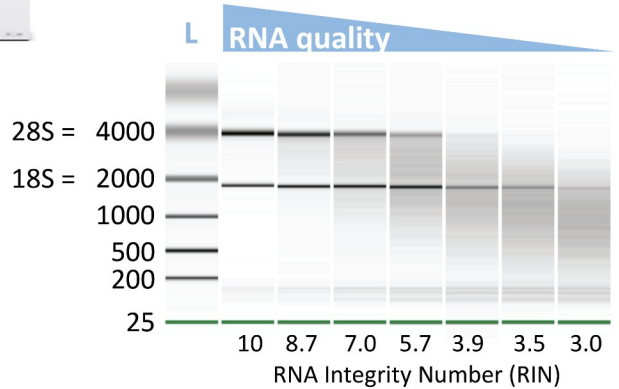
Key point:

✓ xNA quality is crucial

- Quality control:
  - Quantity (ng/µl):
    - Qubit: RNA or DNA High Sensitivity (HS) Assay Kit (ThermoFisher)
  - Quality/ Integrity (length):
    - Capillary electrophoresis
      - Tapestation (Agilent)
      - High Sensitivity DNA or RNA ScreenTape
    - Gel electrophoresis
      - Regular agarose gel for DNA
      - TBE-urea gel for RNA



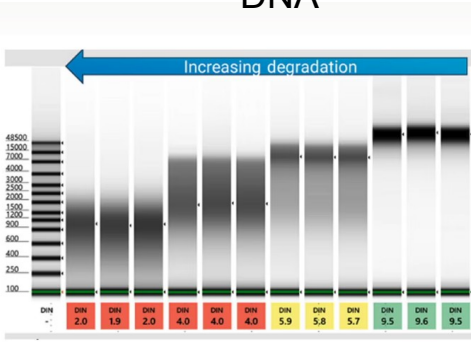




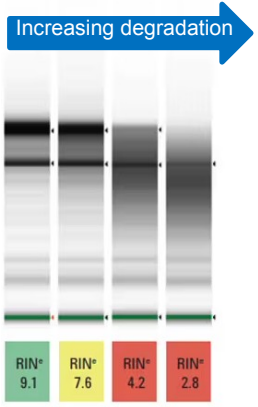
Quality control of extraction: Integrity

Capillary electrophoresis gives a value to assess integrity: a DIN (DNA integrity number) and a RIN (RNA integrity number)

DNA



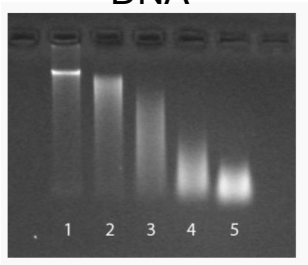
RNA



As RNA degrades the 28S and 18S start to disappea

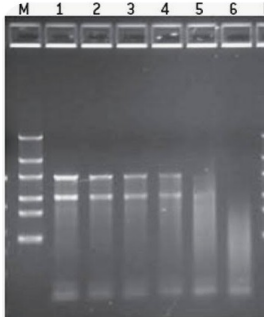
Agarose gel visualization

DNA



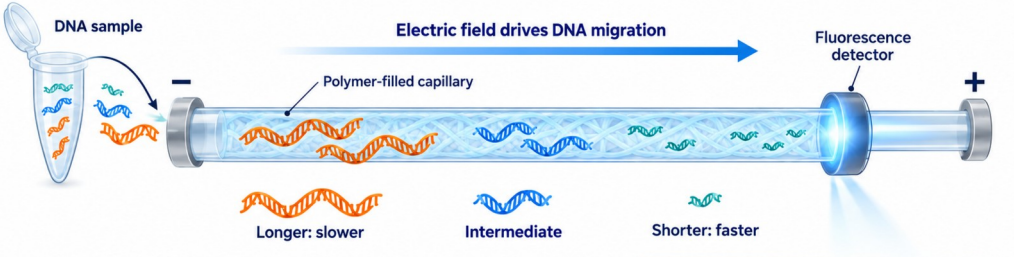
Increasing degradation

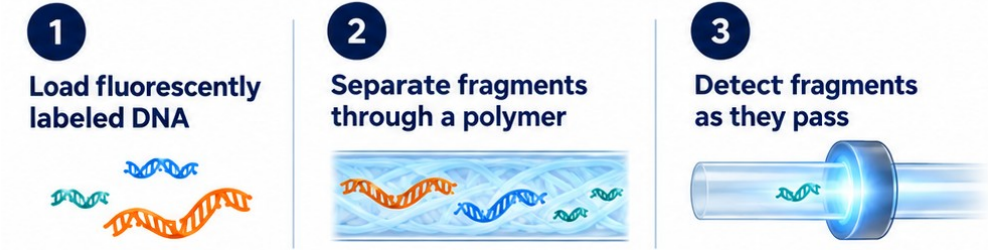
RNA

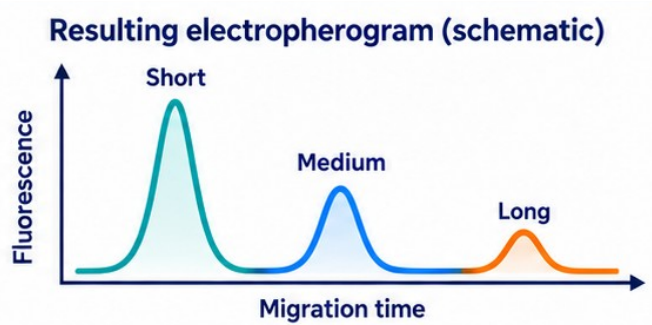


Often low-input samples such as plasma or CSF are below quantification and visualization thresholds

Capillary electrophoresis also separates xNA by size





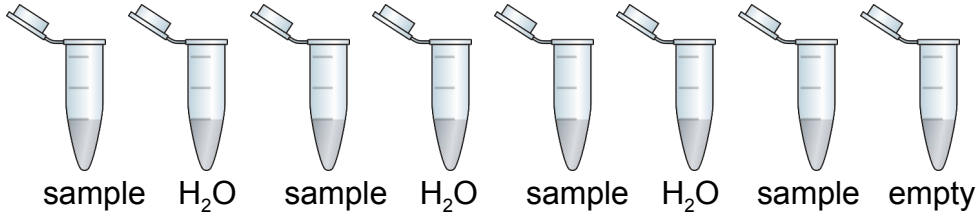


Smaller fragments arrive first and fluorescence produces peacks

xNA extraction

Keep in mind:

- Magnetic stands for 1.5 ml tubes usually accommodate only 8 tubes at a time
- Cross-contamination should be avoided as much as possible!!
  - Process samples with high pathogen content separately from samples with low pathogen content (if you can)
  - Use water controls to estimate the level of cross-contamination
    - Space out samples with water controls, and/or empty spots in the magnetic rack
    - E.g. for 4 samples:



Next: 04-I_Library prep for Illumina / 04-O_Library prep for ONT